Proc. of the 7th International Conference on Engineering and Emerging Technologies (ICEET)
27-28 October 2021, Istanbul, Turkey

HyperExtended LightFace: A Facial Attribute Analysis Framework

Sefik Ilkin Serengil
Applied Data Science Department
Yapi Kredi Teknoloji
Istanbul, Turkey
sefik.serengil@ykteknoloji.com.tr

Alper Ozpinar
Mechatronics Engineering Department
Istanbul Ticaret University
Istanbul, Turkey
aozpinar@ticaret.edu.tr

***Abstract*— Facial attribute analysis from facial images has always been a challenging task. Its practical use cases are very different. This paper mentioned how to build machine learning models with facial image data for age, gender, facial expression, and race/ethnicity prediction tasks. A fully coded open-source software framework was also developed and published with those functionalities.**

Keywords*— *age prediction; gender prediction; emotion prediction; race prediction

I. Introduction

Analyzing facial attributes from facial images has always been a challenging task. Its practical use cases are customer loyalty, marketing and entertainment [1]; kinship recognition such as investigation cases, human trafficking issues, missing children and refugee crises [2]. Even though DNA testing makes kin relationship problems predictable, this is a very expensive and time-consuming process. On the other hand, image-based data is easy to access and process. Moreover, it could contribute to genealogy research because genetic factors in racial and ethnic differences in health and disease are focused intense scrutiny [3]. Furthermore, facial attribute analysis offers an unbiased dataset for facial recognition studies. Herein, demographic information supports those use cases as an important indicator. So, a facial attribute analysis framework developed for age, gender, emotion, and race/ethnicity prediction tasks and explained the timeline and stages of the library in this paper.

II. Related Work

LightFace is a lightweight deep face recognition library for Python [4], a fully open-source software framework licensed under MIT [5]. Its face recognition module wraps several state-of-the-art face recognition models: VGG-Face [6], FaceNet [7], OpenFace [8], DeepFace [9], DeepID [10], Dlib [11] and ArcFace [12]. Those models already reached and passed the human level accuracy when tested on labeled faces in the wild dataset [13].

Besides, detection and alignment are very important early stages of a modern face recognition pipeline. Experiments show that those pre-processing stages improve the accuracy by almost 1% [7]. The library covers many facial detectors: OpenCL OpenCV [14], SSD [15], MTCNN [16], DLIB [11] and RetinaFace [17]. Here, they are all deep learning-based detectors except OpenCV. Besides, MTCN and RetinaFace find facial landmarks including eye, nose and mouth coordinates as well. In this way, detected faces can be aligned more robustly.

So, the facial attribute analysis module was added to LightFace. It uses the same pre-processing functionalities as its face recognition module. This module covers age, gender, emotion and race/ethnicity prediction. Even though the face recognition model wraps pre-trained state-of-the-art models, models were trained for facial attribute analysis. A base VGG-Face model is built with its pre-trained weights because of its simplicity and accuracy [18]; modified for output layer for the custom task; freeze early layers to enable transfer learning for the model was constructed and trained.

III. Datasets

Age, gender and race prediction models are heavily inspired by the DEX model [19]. They shared the pre-trained models for age and gender prediction in Caffe [20]. IMDB and WIKI datasets in that study were also used to train the age and gender models with TensorFlow [21] because of its high adoption in the open-source community. There are 523K instances in those datasets. The face-only version size of the IMDB dataset is 7GB, whereas WIKI dataset is 1GB.

Both IMDB and WIKI datasets are not labeled for race and emotion. Resultantly FairFace [22] dataset was used to train the race prediction model. This has 86K train and 11K test instances. Its target labels are Asian (14.10%), Black (14.16%), Indian (14.20%), Latino-Hispanic (15.40%), Middle-Eastern (10.62%) and White (19.05). As an alternative, UTKFace [23] comes with age, gender and race labels with 10K instances but its labels for the race are Asian, Indian, Black, White and Others. In other words, if FairFace and UTKFace datasets were combined, Latino-Hispanic and Middle-Eastern races should be labeled as others. Thereupon UTKFace dataset was discarded.

Similarly, the FER-2013 [24] dataset was used to train the emotion model. There are 28K train instances and 3.5K test instances in the dataset.

IV. Model Structure

The age, gender and race prediction model were built on the base VGG-Face model, as mentioned in Fig 1. On the other hand, the original study uses regular VGG architecture. VGG-Face expects 224x224x3 shaped inputs, and it has a 22 deep layered architecture. So, the input layer shape is equal to the expected size of VGG-Face in those models. However, the output layers were customized based on the problem. Firstly, the output layer of the gender prediction model is the two correspondings to man and woman. Secondly, the age model has 101 nodes in the output layer corresponding to

2021 International Conference on Engineering and Emerging Technologies (ICEET) | 978-1-6654-2714-2/21/$31.00 ©2021 IEEE | DOI: 10.1109/ICEET53442.2021.9659697

978-1-6654-3897-1/21/$31.00 ©2021 IEEE

Authorized licensed use limited to: McGill Libraries. Downloaded on March 07,2026 at 21:00:45 UTC from IEEE Xplore. Restrictions apply.

ages 0 to 100. Thirdly, the race model has six outputs: Asian, Black, Indian, Latino-Hispanic, Middle-Eastern and White.

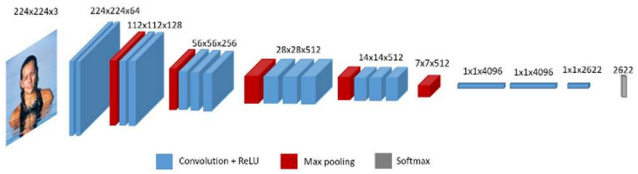

Fig 1. VGG-Face structure

As an exception, the emotion model is not built on VGG-Face structure. Because FER-2013 dataset has 48x48 shaped grayscale images, whereas the other data sets have higher resolution and RGB channels, it has a 12 layered architecture, as mentioned in Table I. There are five convolution layers and three fully connected layers. Its output layer has seven nodes corresponding: Angry, Disgust, Fear, Happy, Sad, Surprise, and Neutral.

TABLE I. EMOTION MODEL STRUCTURE

	1 conv	2 mpool	3 conv	4 conv	5 apool	6 conv
Filters	64	-	64	64	-	128
Kernel	5	-	3	3	-	3
Pool	-	5	-	-	3	-
Strides	-	2	-	-	2	-
Units	-	-	-	-	-	-
	7 conv	8 apool	9 fc	10 fc	11 fc	12 softmax
Filters	128	-	-	-	-	1
Kernel	3	-	-	-	-	-
Pool	-	3	-	-	-	-
Strides	-	2	-	-	-	1
Units	-	-	1024	1024	7	0

V. EVALUATION

Age prediction is a regression task. However, the DEX [19] study authors defined that problem as a classification task and calculated the probability scores for each candidate's age in the output layer. In a regular classification task, the maximum probability score was assigned in the output, but they applied a different approach. Each output score will be multiplied by the corresponding age. In this way, they can predict the apparent age, as shown in Fig. 2.

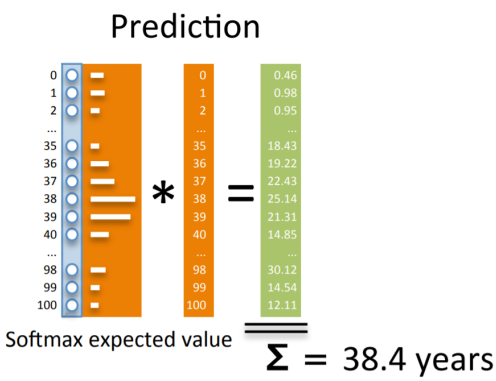

Fig. 2. Apparent Age Prediction Approach *[19]*

The same procedures for the age prediction task [25] were applied. MAE 4.65 was obtained with the test set. In other words, age can be predicted with plus and minus 4.65 years. The raw dataset was split into train, validation and test randomly.

The original study tested its model on ChaLearn Looking at People (LAP) challenge 2015 (ICCV '15) [26]. This challenge has a dataset consisting of 1079 instances with several age predictions of a jury instead of actual ages. Due to this fact, predictions will be evaluated with the following formula considering the mean and standard deviation of the jury predictions.

$$\varepsilon = 1 - e^{-\frac{(x-\mu)^2}{2\sigma^2}} \tag{1}$$

Equation (1) offers a customized loss function. Epsilon closes to 0 if the users prediction is close to the mean of the jury predictions. Similarly, epsilon closes to 0 if users' prediction is not close to the mean of jury predictions but the standard deviation of the jury predictions is high. On the other hand, epsilon becomes bigger if user's prediction is not close to the mean of jury predictions and the standard deviation of jury prediction is low as well.

The epsilon value of our model is 0.387378. On the other hand, the epsilon value of the original study was 0.26975. They also declared that the human-level score is 0.34 [19]. So, the original study is still a little bit more accurate than the proposed model; however a similar score was obtained in the human-level accuracy.

In contrast to age prediction task, gender, race/ethnicity and emotion models are classification models. Evaluating classification models requires some additional metrics such as precision and recall.

$$\text{precision} = TP / (TP + FP) \tag{2}$$

$$\text{recall} = TP / (TP + FN) \tag{3}$$

Equation (2) finds the precision and it focuses on the how many of predictions of a custom class are really true. It is the ratio of true positives to sum of true and false positives. Similarly, (3) finds the recall and it mentions how many of a target class correctly predicted. It is the ratio of true positives to sum of true positives and false negatives.

Gender prediction model has 97.44% accuracy on the same test set with age prediction. Besides, its precision and recall scores for each target class are shown in Table II as well.

TABLE II. CONFUSION MATRIX FOR GENDER MODEL

		prediction		
		Woman	Man	Recall
actual	Woman	1873	98	95,03%
	Man	72	4604	98,46%
	Precision	96,30%	97,92%	

Secondly, the race/ethnicity model has 68% accuracy [27]. Its confusion matrix and corresponding precision-recall values for each target class are demonstrated in Table III. Finally, the emotion model has 57.42% accuracy [28]. Its confusion matrix and precision/recall scores for each target class are shown in Table IV.

Authorized licensed use limited to: McGill Libraries. Downloaded on March 07,2026 at 21:00:45 UTC from IEEE Xplore. Restrictions apply.

TABLE III. CONFUSION MATRIX FOR RACE MODEL

		prediction						
		Asian	Indian	Black	White	Middle Easter	Latino Hispan	Recall
actual	Asian	2683	63	39	81	10	89	90%
	Indian	116	1209	74	112	8	37	78%
	Black	151	93	817	285	88	82	54%
	White	237	74	111	735	136	330	45%
	Middle Eastern	45	17	71	241	500	335	41%
	Latino Hispanic	140	20	17	232	152	1524	73%
	Precision	80%	82%	72%	44%	56%	64%	

TABLE IV. CONFUSION MATRIX FOR EMOTION MODEL

		prediction							
		Angry	Disgust	Fear	Happy	Sad	Surprise	Neutral	Recall
actual	Angry	214	9	53	30	67	8	86	46%
	Disgust	10	24	9	2	6	0	5	43%
	Fear	45	2	208	29	89	45	78	42%
	Happy	24	0	40	696	37	18	80	78%
	Sad	65	3	83	56	285	10	151	44%
	Surprise	7	1	42	27	9	303	26	73%
	Neutral	45	2	68	65	88	8	331	55%
	Precision	52%	59%	41%	77%	49%	77%	44%	

VI. SOFTWARE FRAMEWORK

Experiments show that facial attribute analysis, including age, gender, emotion and race, can be predicted robustly. All these functionalities were planned to design as a pipeline. It begins with detecting faces and aligns it with clearing the noise and having a more robust result. As mentioned, the LightFace face recognition module already wraps several advanced detectors. So, the facial attribute analysis service expects an optional detector name. Its default parameter values were set as OpenCV [14] because of its high speed. Still, RetinaFace [17], SSD [15], MTCNN [16] and DLIB [11] could be applicable.

Then, it is going to build the base VGG-Face model first and customize it based on the attribute type. Here, VGG-Face has a simple structure and it could be built in a few lines of code. Consequently, the model structure is going to be stored in the code repository. On the other hand, that model has millions of weight parameters. Its pre-trained weight file is about 500MB. Duplicating it to age, gender and race model requires almost 1.5GB of storage. Thence, storing pre-trained weights in the code repository is not applicable. The proposed framework and model default settings store those pre-trained weights in GitHub as a source and download them once in the production environment.

So, future users easily download the pre-trained weights from an external source if required, load the pre-trained weights, and apply pre-processing for images, including detection and alignment in the background. Users should indicate the exact path of an image to the facial attribute service of LightFace framework, as shown in Table V. Then, it is going to analyze age, gender, emotion and race.

TABLE V. CODE SNIPPET FOR ANALYSIS FUNCTIONALITY

```
#!pip install deepface
from deepface import DeepFace

backends = ["opencv", "ssd", "mtcnn", "dlib", "retinaface"]

obj = DeepFace.analyze("img.jpg"
    , actions = ["age", "gender", "emotion", "race"]
    #, detector_backend = backends[0]
)
```

Analysis functionality returns the facial attribute analysis results in json format as shown in Fig. 3.

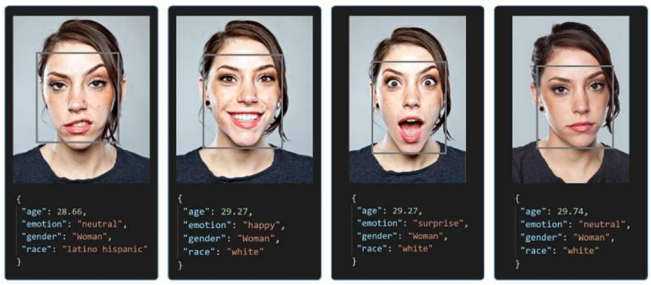

Fig. 3. Result of Analysis Functionality

VII. CONCLUSION

So, a facial attribute analysis module was also added into the existing deep face recognition framework LightFace. Because face recognition and facial attribute analysis are highly related and the hyper extended framework will contribute to the common knowledge in face detection and processing algorithms. This paper explains the stages of the development of this module as well as all technical details of the model building and training stages. LightFace, with its facial attribute analysis module, can contribute the solution to many problems such as unbiased facial recognition training, marketing, kinship recognition or genealogy researches.

REFERENCES

[1] A. Herruzo and N. Pashenkov, "Collection to Creation: Playfully Interpreting the Classics with Contemporary Tools" in *The International Conference on Computational Design and Robotic Fabrication*, 2020.

[2] J. P. Robinson, H. L. Yue Wu and Y. Fu, "Visual kinship recognition of families in the wild" *IEEE Transactions on pattern analysis and machine intelligence,* vol. 40, no. 11, 2018.

[3] R. Bulatao and N. Anderson, "Understanding Racial and Ethnic Differences in Health in Late Life: A Research Agenda" in *National Academies Press (US)*, 2004.

[4] S. I. Serengil and A. Ozpinar, "LightFace: A Hybrid Deep Face Recognition Framework" in *2020 Innovations in Intelligent Systems and Applications Conference (ASYU)*, 2020.

[5] S. I. Serengil, "DeepFace: A Lightweight Deep Face Recognition and Facial Attribute Analysis (Age, Gender, Emotion and Race) Framework for Python" [Online]. Available: https://github.com/serengil/deepface.

[6] O. M. Parkhi, A. Vedaldi and A. Zisserman, "Deep face recognition" 2015.

Authorized licensed use limited to: McGill Libraries. Downloaded on March 07,2026 at 21:00:45 UTC from IEEE Xplore. Restrictions apply.

[7] F. Schroff, D. Kalenichenko and J. Philbin, "Facenet: A unified embedding for face recognition and clustering" in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2015.

[8] T. Baltrušaitis, P. Robinson and L.-P. Morency, "Openface: an open source facial behavior analysis toolkit" in *2016 IEEE Winter Conference on Applications of Computer Vision (WACV)*, 2016.

[9] Y. Taigman, Yaniv, M. Yang, M. Ranzato and L. Wolf, "Deepface: Closing the gap to human-level performance in face verification" in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2014.

[10] Y. Sun, D. Liang, X. Wang and X. Tang, "Deepid3: Face recognition with very deep neural networks" in *arXiv preprint arXiv:1502.00873*, 2015.

[11] D. E. King, "Dlib-ml: A machine learning toolkit." *The Journal of Machine Learning Research,* vol. 10, pp. 1755-1758, 2009.

[12] J. Deng, J. Guo, N. Xue and S. Zafeiriou, "Arcface: Additive angular margin loss for deep face recognition" in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019.

[13] G. B. Huang, M. Mattar, T. Berg and E. Learned-Miller, "Labeled faces in the wild: A database forstudying face recognition in unconstrained environments" in *Workshop on faces in'Real-Life'Images: detection, alignment, and recognition*, 2008.

[14] G. Bradski, "The opencv library" *Dr Dobb's J. Software Tools,* vol. 25, pp. 102-125, 2000.

[15] W. Liu, D. Anguelov, D. Erhan, C. Szegedy, S. Reed, C.-Y. Fu and A. C. Berg, "SSD: Single Shot MultiBox Detector" *Computer Vision – ECCV 2016. ECCV 2016. Lecture Notes in Computer Science,* vol. 9905, 2016.

[16] J. Xiang and G. Zhu, "Joint face detection and facial expression recognition with MTCNN" in *4th International Conference on Information Science and Control Engineering (ICISCE). IEEE*, 2017.

[17] J. Deng, J. Guo, Y. Zhou, J. Yu, I. Kotsia and S. Zafeiriou, "Retinaface: Single-stage dense face localisation in the wild" in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020.

[18] S. I. Serengil, "Deep Face Recognition with Keras" [Online]. Available: https://sefiks.com/2018/08/06/deep-face-recognition-with-keras/.

[19] R. Rothe, R. Timofte and L. V. Gool, "Deep expectation of real and apparent age from a single image without facial landmarks." *International Journal of Computer Vision,* vol. 126, no. 2, pp. 144-157, 2018.

[20] Y. Jia, E. Shelhamer, J. Donahue, S. Karayev, J. Long, R. Girshick, S. Guadarrama and T. Darrell, "Caffe: Convolutional Architecture for Fast Feature Embedding" in *arXiv preprint arXiv:1408.5093*, 2014.

[21] M. Abadi, "Tensorflow: A system for large-scale machine learning" in *12th {USENIX} symposium on operating systems design and implementation*, 2015.

[22] K. Kärkkäinen and J. Joo, "FairFace: Face Attribute Dataset for Balanced Race, Gender, and Age for Bias Measurement and Mitigation" in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2021.

[23] Z. Zhang, Y. Song and H. Qi, "Age progression/regression by conditional adversarial autoencoder" in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2017.

[24] I. J. Goodfellow, "Challenges in Representation Learning: A report" in *International conference on neural information processing*, 2013.

[25] S. I. Serengil, "Apparent Age and Gender Prediction in Keras" [Online]. Available: https://sefiks.com/2019/02/13/apparent-age-and-gender-prediction-in-keras/.

[26] S. Escalera, X. Bar´o and J. Gonz`alez, "ChaLearn Looking at People Challenge 2014:" in *European Conference on Computer Vision*, 2014.

[27] S. I. Serengil, "Race and Ethnicity Prediction in Keras" [Online]. Available: https://sefiks.com/2019/11/11/race-and-ethnicity-prediction-in-keras/.

[28] S. I. Serengil, "Facial Expression Recognition with Keras" [Online]. Available: https://sefiks.com/2018/01/01/facial-expression-recognition-with-keras/.

Authorized licensed use limited to: McGill Libraries. Downloaded on March 07,2026 at 21:00:45 UTC from IEEE Xplore. Restrictions apply.